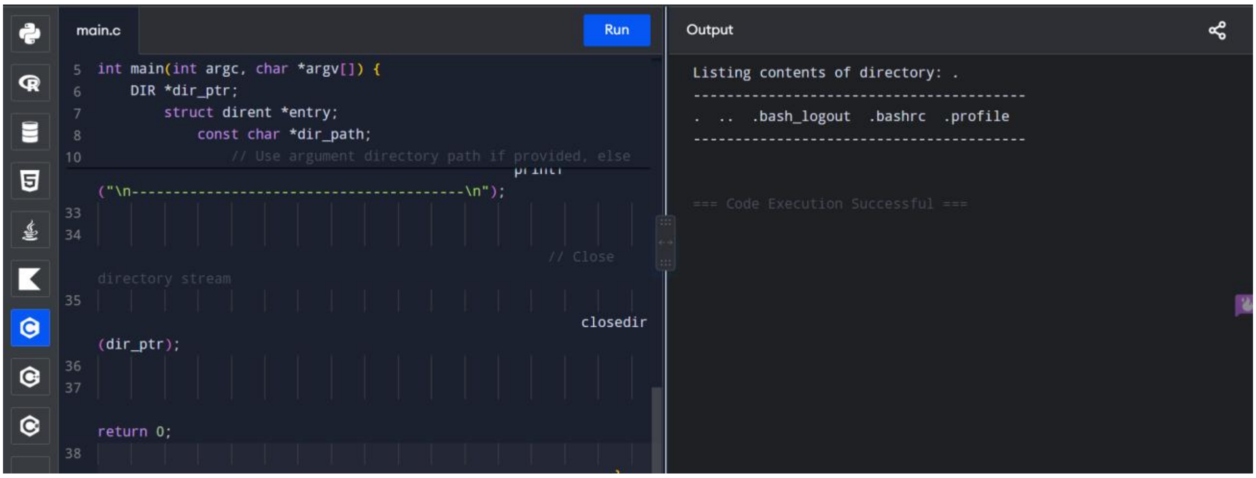
EXPERIMENT 37:

37. Develop a C program for simulating the function of ls UNIX Command.



PSEUDO CODE :

- Check if a directory path argument was provided via command-line arguments (argc, argv). If provided, set target directory to argv[1]; otherwise, default to standard current directory ".".
- Open the directory stream using opendir().
- If directory pointer is NULL, print an error message using perror() and exit.
- Loop through the directory entries using readdir() inside a while loop.
- In each iteration, retrieve the filename string from entry->d_name and display it.
- Close the directory stream using closedir().

CODE:

```
#include <stdio.h>

#include <stdlib.h>

#include <dirent.h>
```

```
int main(int argc, char
*argv[]) {
    DIR *dir_ptr;
    struct dirent *entry;
    const char *dir_path;


    // Use argument
directory path if
provided, else default to
current directory "."
    if (argc > 1) {
        dir_path = argv[1];
    } else {
        dir_path = ".";
    }


    // Open directory
stream
    dir_ptr =
```

```
opendir(dir_path);

    if (dir_ptr == NULL)
{

        perror("Unable to
open directory");

        return 1;

    }



    printf("Listing
contents of directory:
%s\n", dir_path);

    printf("-----------------
-----------------------\n");



    // Read and print
directory entries

    while ((entry =
readdir(dir_ptr)) !=
NULL) {

        // Print filename
```

```
        printf("%s  ",
entry->d_name);

    }

    printf("\n---------------
-------------------------
\n");



    // Close directory
stream

    closedir(dir_ptr);



    return 0;

}
```

INPUT :

./ls_sim

OUTPUT:

Listing contents of directory: .

. .. main.c ls_sim

Makefile notes.txt

data.csv
